Current + Past Papers Of BT-603 (Fermentation Technology) Fall-2022

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Enlist six trace elements used in fermentation.

Trace elements are essential nutrients required by microorganisms for their growth and metabolism during fermentation. Here are six trace elements commonly used in fermentation:

- Iron (Fe)
- Zinc (Zn)
- Manganese (Mn)
- Copper (Cu)
- Cobalt (Co)
- Molybdenum (Mo)

Write down micro and macro minerals used in fermentation.

Minerals play an essential role in fermentation, both at the micro and macro levels. Micro minerals, such as iron, zinc, and copper, act as cofactors for enzymes involved in fermentation, allowing for the proper functioning of metabolic processes. Macro minerals, such as calcium and magnesium, are important for maintaining pH balance and providing structural support for microorganisms during fermentation. Overall, minerals are crucial for optimal fermentation performance and the production of high-quality fermented products.

Method of antibiotic screening.

The method of antibiotic screening typically involves the following steps:

- Collection of microorganisms: A variety of microorganisms, including bacteria and fungi, are collected from different sources such as soil, water, and plants.
- Isolation of strains: Individual strains of microorganisms are isolated and grown in culture.
- Testing for antibiotic production: The isolated strains are tested for their ability to produce antibiotics using various methods such as the agar well diffusion assay, disk diffusion assay, and broth dilution assay.
- Identification of active compounds: The active compounds produced by the microorganisms are identified using techniques such as chromatography and spectrophotometry.
- Characterization of active compounds: The active compounds are further characterized for their chemical and physical properties, as well as their efficacy against specific bacterial or fungal strains.
- Optimization of production: The conditions for optimal production of the antibiotic are determined, and the yield of the active compound is increased.

Overall, antibiotic screening involves a combination of microbiological, biochemical, and chemical techniques to identify and characterize new antibiotics.

Differentiate between aerobic and anaerobic fermentation.

Aerobic fermentation requires oxygen, while anaerobic fermentation occurs in the absence of oxygen. During aerobic fermentation, glucose is broken down completely into carbon dioxide and water, producing a large amount of energy in the form of ATP.

Anaerobic fermentation involves the breakdown of glucose into smaller molecules such as lactic acid or ethanol, which produces less ATP. Anaerobic fermentation is commonly used in the production of certain foods and beverages such as yogurt, cheese, and beer.

How oxygen is supplied to the microbial culture during fermentation.

During fermentation, oxygen is supplied to microbial cultures through various methods such as agitation, sparging, or the use of oxygen-enriched air. Agitation involves stirring or shaking the culture to increase the contact between the microbial cells and the air. Sparging involves bubbling air or oxygen into the culture to provide a continuous supply of oxygen. Oxygen-enriched air can also be used to supply oxygen to the culture. The method used depends on the specific fermentation process and the type of microorganisms being cultured.

OR

Oxygen is normally supplied to microbial cultures in the form of air, this being the cheapest available source of the gas. The method for provision of a culture with a supply of air varies with the scale of the process:

- **Laboratory-scale cultures** may be aerated by means of the shake-flask technique where the culture (50 to 100 cm3) is grown in a conical flask (250 to 500 cm3) shaken on a platform contained in a controlled environment chamber.
- **Pilot- and industrial-scale fermentations** are normally carried out in stirred, aerated vessels (fermenters).

Write and explain stoichiometric equation.

Stoichiometric equations are balanced chemical equations that represent the relationship between the reactants and products in a chemical reaction. The coefficients in the equation represent the number of molecules, atoms, or ions involved in the reaction. For example, the stoichiometric equation for the combustion of methane (CH4) in the presence of oxygen (O2) is:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

This equation shows that one molecule of methane reacts with two molecules of oxygen to produce one molecule of carbon dioxide and two molecules of water. The stoichiometric equation is important because it allows us to determine the amount of reactants needed to produce a certain amount of product, and vice versa. It also helps us to understand the chemical properties of a substance and the nature of the chemical reactions that it undergoes.

Explain biological and physical problems due to formation of excessive foam in fermentation.

Excessive foam formation during fermentation can cause both biological and physical problems. Biologically, the foam can inhibit the exchange of gases between the fermentation broth and the surrounding air, which can impair the growth and metabolism of microorganisms involved in the fermentation process. Physically, the foam can cause spills, overflows, and blockages in equipment and pipes, which can disrupt the operation of the fermentation process and lead to contamination and product loss.

Steps of introducing air into liquid?

The steps for introducing air into a liquid can be summarized as follows:

- **Select a method of aeration:** There are different ways to introduce air into liquids, including mechanical agitation, sparging, and diffusion.
- **Determine the desired level of aeration:** The amount of air introduced into the liquid will depend on the specific application and desired outcome.
- **Prepare the liquid:** The liquid may need to be degassed or filtered before aeration to ensure optimal results.
- **Introduce air into the liquid:** Depending on the method chosen, air can be introduced by bubbling it through the liquid, spraying it into the liquid, or stirring the liquid vigorously.
- **Monitor and control the process:** It is important to monitor the process to ensure the desired level of aeration is achieved and to make adjustments if necessary.
- **Allow for settling:** After introducing air into the liquid, it may need to settle for a while to allow for any bubbles to rise to the surface.
- **Use the aerated liquid as desired:** Once the aeration process is complete, the liquid can be used for its intended purpose.

Steps Bartholomew for oxygen introduction into liquid medium?

The Steps Bartholomew method for introducing oxygen into a liquid medium involves the following steps:

1. Fill a sterile flask or container with the liquid medium.
2. Inoculate the medium with the desired microorganisms.
3. Attach a sterile air sparger to the bottom of the flask or container.
4. Connect the air sparger to an air pump or compressed air source.
5. Adjust the airflow to provide sufficient oxygenation for the microorganisms to grow.
6. Incubate the flask or container under the desired conditions, monitoring the growth of the microorganisms.

Overall, the Steps Bartholomew method is a common and effective way to introduce oxygen into a liquid medium for the growth of microorganisms.

Enlist any five names of industrially important enzymes inducers that used in fermentation?

1. Starch Maltose
2. Maltose
3. Yeast mannans
4. Phenyl acetic acid Various proteins
5. Cellulose
6. Pectin (beet pulp, apple pomace, citrus peel)
7. Isovaleronitrile

Enlist any five names of Inhibitors that used in fermentation?

1. Ethanol
2. Lactic Acid
3. Sulphur Dioxide
4. Antibiotics
5. Carbon Dioxide
6. Sodium chloride
7. Sodium bisulphate
8. Bromide
9. Penicillin
10. Alkali metal/Phosphate, pH below 2.0
11. Di-ethyl barbiturate

Write the names of the institutes of the cultures collection?

- American Type Culture Collection (ATCC)
- National Collection of Industrial and Marine Bacteria (NCIMB)
- Culture Collection of Algae and Protozoa (CCAP)
- Deutsche Sammlung von Mikroorganismen und Zellkulturen (DSMZ)
- Japanese Collection of Microorganisms (JCM)
- National Collection of Type Cultures (NCTC)
- Leibniz Institute DSMZ – German Collection of Microorganisms and Cell Cultures
- Collection of Industrial Microorganisms (CIM)
- Belgian Co-ordinated Collections of Micro-organisms (BCCM)
- China General Microbiological Culture Collection Center (CGMCC)

Basic component of fermentation? 3M

The basic components of fermentation are:

- Microorganisms (such as yeast, bacteria or fungi)

- A substrate (such as sugar or starch)
- Anaerobic (oxygen-free) conditions

During fermentation, microorganisms metabolize the substrate, producing energy and various by-products, such as alcohol or lactic acid.

Describes the streak plate method? 3M

The streak plate method is a microbiological technique used to isolate and purify colonies of bacteria from a mixed sample. It involves using a sterile inoculating loop to streak a small amount of the sample onto a solid agar plate in a pattern that progressively dilutes the bacteria. This results in the separation of individual colonies that can be further analyzed or used for experiments.

Give criteria Bull et al. (1979) as being important in the choice of process organism? 5M

Bull et al. (1979) identified four criteria as important in the choice of process organism:

- The organism should have a high growth rate to maximize productivity.
- The organism should be able to utilize the substrate efficiently.
- The organism should produce the desired product with high yield.
- The organism should be amenable to genetic manipulation for strain improvement.

What factors affect the rate of diffusion of oxygen through the boundary layer?

The factors that affect the rate of diffusion of oxygen through the boundary layer include

- Temperature
- Pressure
- Concentration gradient
- Surface area
- The nature of the boundary layer

Three physical mutagens?

Three physical mutagens are:

1. UV light
2. Ionizing radiations

- X-Rays
- Gamma Rays
- Cosmic Rays
- Alpha Rays
- Fast neutrons

Types of fermentation on scale?

There are two main types of fermentation on a large scale:

- **Batch Fermentation:** In batch fermentation, all the nutrients required for fermentation are added at the beginning of the process, and the fermentation continues until the substrate is consumed or the desired product is formed.
- **Continuous Fermentation:** In continuous fermentation, a constant flow of substrate is added to the fermentation vessel, and the product is continuously removed. This allows for a steady-state operation and higher product yields.

Storage under Liquid Nitrogen?

Liquid nitrogen should be stored in a well-ventilated, insulated container that is specifically designed for cryogenic liquids. The container should be made of materials that can withstand extreme cold temperatures and should be kept in a cool, dry, and well-ventilated area. It is important to follow all safety precautions when handling and storing liquid nitrogen, as it can be extremely cold and potentially dangerous.

OR

The metabolic activities of micro-organisms may be reduced considerably by storage at the very low temperatures (-150° to -196°) which may be achieved using a liquid nitrogen refrigerator.

Method to manipulate genetic material?

One method to manipulate genetic material is genetic engineering, which involves the use of techniques to introduce, remove, or modify genes in an organism's genome. This can be achieved through various methods such as gene editing using CRISPR-Cas9, gene transfer using viral vectors to modify or manipulate genes or genomes, gene synthesis, recombinant DNA technology and gene editing.

Antifoam three ways to control?

Antifoam is a substance used to control foam in various industrial processes. Here are three ways to control foam using antifoam:

- **Addition:** One way to control foam is by adding antifoam directly into the foaming solution. This helps to break down the foam and prevent it from forming.
- **Dilution:** Another method is to dilute the foaming solution with water or another liquid. This reduces the concentration of the foam-forming agents and can help prevent foam from forming.
- **Surface Tension Reduction:** Antifoam agents can also reduce the surface tension of the liquid, which can help prevent the formation of foam. This method is often used in applications such as wastewater treatment, where foaming can be a problem.

Negative and Positive Selection? 3 Marks

Positive selection involves the selection of mutant cells and rejection of non- mutated cells. To identify mutants that have gained some physical trait,

Negative selection identifies mutants that have lost some physical trait – for example, auxotrophs that have nutritional requirements not possessed by the parent (non-mutated) cell. Replica plating is used for negative selection.

Bio fertilizers? 3 Marks

Bio fertilizers are natural fertilizers that contain living microorganisms such as bacteria, fungi, or algae. These microorganisms help in the growth and nutrient uptake of plants by increasing the availability of nutrients such as nitrogen, phosphorus, and potassium in the soil. They are an environmentally friendly and sustainable alternative to chemical fertilizers.

Limitations of starch? 3 Marks

The main limitations of starch include:

- Limited solubility in cold water
- Limited ability to thicken acidic solutions
- Limited stability under high temperatures and acidic conditions
- Limited compatibility with certain food ingredients and processing techniques

Difference between Agitation and Aeration?

Agitation refers to the process of stirring or mixing a substance using mechanical means, such as a mixer or a stirrer. Agitation can be used to promote the even distribution of substances, increase reaction rates, or prevent settling.

Aeration refers to the process of introducing air or oxygen into a substance or environment. Aeration can be used to oxygenate water in aquariums or ponds, improve soil quality, or promote the aerobic decomposition of organic matter.

In summary, **agitation** involves mixing or stirring, while **aeration** involves introducing air or oxygen.

Explain CPR? 5 Marks

Parton and Willis (1990) advocated the use of the carbon dioxide production rate (CPR) as a transfer criterion which requires analysis of the fermenter effluent air. This approach is suitable only when transfer is being made from a fermenter. These workers provide an excellent example of the effect of inoculum transfer time on the productivity of a Streptomyces secondary metabolite. The carbon dioxide production rate (CPR) profile of the inoculum culture showing the points 0, 2 and 3) at which inocula were removed. The effect of inoculum age on the CPR of the production fermentation, it may be seen that the three fermentations performed similarly. Thus, although the time of transfer had only a marginal influence on biomass in the production fermentation the effect on product formation was critical.

Formation of Protoplasts? 2 Marks

Protoplasts are formed from bacteria, fungi, yeast and actinomycetes when dividing cells are caused to lose their cell walls.

Problems in Medium optimization? 2 Marks

Medium optimization by the classical method of changing one independent variable (nutrient, antifoam, pH, temperature, etc.) while fixing all the others at a certain level can be extremely time consuming and expensive for a large number of variables.

Chemical Mutagens? 3 Marks

Chemical mutagens are substances that can cause changes or mutations in the genetic material of an organism, such as DNA. These mutations can lead to altered or abnormal functioning of cells, and potentially cause diseases such as cancer. Examples of chemical mutagens include certain types of pesticides, industrial chemicals, and tobacco smoke.

Steps involved in transferring oxygen from air to cell in fermentation? 3 Marks

Bartholomew et al. (1950) represented the transfer of oxygen from air to the cell, during a fermentation, as occurring in a number of steps:

i. The transfer of oxygen from an air bubble into solution.
ii. The transfer of the dissolved oxygen through the fermentation medium to the microbial cell.
iii. The uptake of the dissolved oxygen by the cell.

Name the compounds used in Antifoaming? 5 marks

- Alcohols; stearyl and octyl decanol
- Esters
- Fatty acids and derivatives, particularly glycerides, which include cottonseed oil, linseed oil, say-bean oil, olive oil, castor oil, sunflower oil, rapeseed oil and cod liver oil
- Silicones
- Sulphonates.
- Miscellaneous; Alkaterge C, oxazaline, poly-propylene glycol

Write two examples of methylotrophic yeast? 3 marks

Methylotrophic yeasts such as Hansenula polymorpha and Pichia pastoris.

Write down the factors affecting KLa value. 3 Marks

Factors affecting KLa the air-flow rate employed in vessels,

- The degree of agitation inside vessels.
- The rheological properties of the culture broth.
- The presence of antifoam agents.

What is Henry's Law? 3 Marks

Henry's Law describes the solubility of O2 in nutrient solution in relation to the O2 partial pressure in the gas phase.

$$C=Po/H$$

- C* is the oxygen saturation concentration of the nutrient solution,
- Po is the partial pressure of the gas in the gas phase
- H is Henry's constant, which is specific for the gas and the liquid phase.

Explain Ames Test. 5 marks

The Ames test is a relatively inexpensive and rapid test for identifying possible chemical carcinogens. The test assumes that a mutant cell can revert to a normal cell in the presence of a mutagen and that many mutagens are carcinogens. Histidine auxotrophs (require histidine in the media) of Salmonella are exposed to an enzymatically treated potential carcinogen, and reversions to the non-mutant state are selected. Before considering the methods used for the selection of methods producing improved levels of primary metabolites it is necessary to study the mechanisms of control of their biosynthesis such that the 'blueprints', may be drawn accurately.

Name the various Phenotypic technique for identification of microorganism? 3 Marks

- Microscopic morphology
- Metabolic differences (Biochemical Testing)
- Serology
- Fatty acid analysis

Methods for genetic technique industrial process. 3 Marks

1. Conventional mutation/Random Mutagenesis
2. Transduction
3. Conjugation
4. Transformation
5. Protoplast fusion
6. Genetic engineering
7. Metabolic engineering

Dynamic and static method is suitable for the determination of K1a value? 3 Marks

Both the dynamic and static methods are also unsuitable for measuring KLa values in viscous systems.

Explain industrial strategy for usage of microorganisms with Flow sheet Diagram? 5 Marks

A. Isolation of microbes from natural source
B. Identification of desired microbes
C. Characterization of microbes
D. Screening of desired microbes
E. Strain Improvement
F. Fermentation

What is the function of Impeller in the fermentation process? 3 Marks

Fermentation involves proper mixing during the process of fermentation which is done by Agitator/ Impeller.

What is the role of inhibitor in fermentation medium? 3 Marks

The role of an inhibitor in a fermentation medium is to limit or prevent the growth and metabolism of unwanted microorganisms, which can compete with the desired microorganism for nutrients and affect the yield and quality of the fermentation process. Inhibitors can also help to maintain the stability and purity of the fermentation culture by reducing the risk of contamination.

What is random mutagenesis? 3 marks

Random mutagenesis is a technique used in molecular biology to introduce random mutations into DNA sequences to generate genetic diversity and study the effects of mutations on protein function.

Phonotypic characteristics of microorganisms? 3 Marks

Phenotypic characteristics of microorganisms refer to their observable physical and biochemical traits, such as size, shape, colour, motility, metabolic capabilities, and the ability to produce certain enzymes or metabolites. These characteristics are important for the identification and classification of microorganisms.

Name the three ways of introducing the air in the fermenter. 3 Marks

The three ways of introducing air in the fermenter are:

Sparging: Introducing air by bubbling it through a sparger at the bottom of the fermenter.

Surface aeration: Introducing air by spraying it over the surface of the liquid in the fermenter.

Mechanical agitation: Introducing air by mechanically agitating the liquid in the fermenter, such as by stirring or shaking.

What is Replica plating Method? 3 Marks

Replica plating is a laboratory technique used to transfer bacteria from one solid growth medium to another solid growth medium while maintaining the spatial arrangement of the colonies. It involves pressing a velvet-covered disk onto a bacterial culture and then using the same disk to transfer bacteria onto a new medium, allowing for the isolation and comparison of different bacterial strains.

Write a note on molasses? 5 Marks

Molasses is a thick, dark, and sweet syrup that is produced during the process of refining sugar cane or sugar beets into sugar. It is commonly used as a sweetener in baking and cooking, as well as in the production of rum and other spirits. Molasses is also a good source of vitamins and minerals, including iron, calcium, and potassium. There are several different types of molasses, including light molasses, dark molasses, and blackstrap molasses, each with their own unique flavour and nutritional profile.

Technique of microbial culture storage/preservation explains it? 5 Marks

Microbial culture storage and preservation techniques involve methods to maintain the viability and genetic stability of microorganisms for extended periods of time. These techniques typically include freezing, lyophilisation (freeze-drying), and storing cultures in liquid nitrogen or glycerol at low temperatures. The choice of technique depends on the type of microorganism and the intended use of the culture. Proper storage and preservation of microbial cultures ensure their availability for future research and applications.

Objective

1. Inhibitors have also been used to affect cell-wall structure and increase the permeability for release of **metabolites**.

2. Supply of oxygen is the rate limiting step In an **aerobic fermentation.**
3. The rate of movement of an oxygen molecule through the bulk liquid is dependent on the degree of mixing (relative to the volume of the reactor) and **viscosity** of the medium.
4. The dissolved oxygen concentration, is zero then the KLa may then be calculated from the equation **OTR = KLa . C***.
5. Early screening strategies tended to be empirical, labour intensive and had relatively **low** success rates.